ELSEVIER

Earth and Planetary Science Letters 125 (1994) 173–191

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Magnetostratigraphy of the Mayerling section (Austria) and Erenkolu Mezarlik (Turkey) section: Improvement of the Carnian (late Triassic) magnetic polarity time scale

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Received 18 January 1994; revision accepted 19 May 1994

Abstract

We present a magnetostratigraphic study of two Carnian (late Triassic) pelagic limestone sections from the Northern Calcareous Alps (Mayerling section) and southwestern Turkey (Erenkolu Mezarlik section). Biostratigraphic control is based on conodont zonation for the Mayerling section, and both on conodonts and ammonoids for the Erenkolu Mezarlik section. In both sections, the characteristic magnetization is essentially carried by a mineral of the magnetite family. Eight magnetic intervals are obtained from the Mayerling section and fourteen from Erenkolu Mezarlik. When these results are compared to the Carnian magnetostratigraphic sequence obtained from the Bolücektasi Tepe section, satisfactory correlations are observed and a composite Carnian magnetic polarity time scale containing 24 intervals can be proposed by combining all these results. The lower part of the upper Carnian is however missing (Tuvalian 1 zone and lower part of the Tuvalian 2 zone). No clear correlation is observed between our composite sequence and the other published magnetostratigraphic data of Carnian age. The satisfactory correlation observed between the Mayerling and the Bolücektasi Tepe sections indicates that both sections were deposited in the northern hemisphere, thus providing constraints on the origin of the Antalya nappes.

1. Introduction

The analysis of changes in magnetic reversal frequency since the end of the Kiaman superchron (Late Permian) requires the filling of wide gaps in the Triassic magnetostratigraphic database. In order to establish a late and middle Triassic magnetic polarity time scale, we investigated the magnetostratigraphy of two late Triassic sections from Turkey, the Carnian to Norian Bolücektasi Tepe section (BT section) [1] and the Norian Kavur Tepe section (KT section) [2]. More recently, magnetostratigraphic results have been obtained for the middle Triassic from two Ladinian sections from the Northern Calcareous Alps [3]. From these results, a magnetic polarity sequence containing more than 60 intervals and well-calibrated with the biostratigraphic zonations (ammonoids and conodonts) has been pro-

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SSDI 0012-821X(94)00117-H

posed for the Ladinian–Norian interval [4]. We roughly estimate a reversal frequency of about 2 reversals/my for the Ladinian, 4 reversals/my for the Carnian and about 3 reversals/my for the Norian. These values are remarkably higher than those derived from magnetostratigraphic data mainly obtained from North America [e.g., 5–8]. In general, large discrepancies are observed with these previous results and complementary results are clearly required to test the accuracy of the suggested magnetic sequences.

We report the magnetostratigraphy of two pelagic limestone sections of Carnian age from the Northern Calcareous Alps (Mayerling section) and from southwestern Turkey (Erenkolu Mezarlik (EM) section). These two sections partially overlap the BT section [1], thus making a comparison possible. With respect to the biostratigraphic constraints, the satisfactory correlation which is observed between the sections improves the reference magnetic polarity time scale for the Carnian. However, no datum is yet availble for the lower part of the upper Carnian because this time interval is usuallly missing in the Tethyan realm. Nevertheless, the magnetostratigraphic correlation between the Mayerling and the BT sections provides important constraints on the origin of the Tauride blocks [2].

2. Carnian biochronology

In the tropical and low-latitude belts of the Tethys area, the Carnian stage is defined by five ammonoid zones and nine subzones (Fig. 1) [9]. The lower Carnian, or Julian substage, contains two ammonoid zones (Julian 1 and 2 zones, including the former Cordevolian substage) [9], rather than the three suggested by Kozur [10]. Three ammonoid zones define the upper Carnian, which is also called the Tuvalian substage (Tuvalian 1–3) [11]. This standard zonation, based on the current range zone status, has been calibrated with conodont and halobiid zones [Krystyn, in prep.]. Ten conodont zones are recognized mostly from rocks in the Hallstatt facies: five in the lower Carnian, and five in the upper Carnian [11,12]. Two of these zones, *mostleri* and *carpathicus*, have been proposed by Kozur [10]. The ranges of platform conodont species and their zonal boundaries are integrated with the ammonoid time scale. The uppermost Julian 1 age of the *Metapolygnatus carnicus* zone has been very recently proven by ammonites (*Trachyceras* n.sp. 1, *Trachyceras* cf. *semiramis*) from the Cassian Formation (Cortina, Upper Italy [L. Krystyn, unpublished data]). The conodont divisions mainly have the status of interval zones (I.Z.), and range zones (R.Z.) are rare. The halobiid zonation is based on many sections from Austria, Italy, Slovakia, Slovenia, Bosnia and Turkey [Krystyn, in prep.]. Although the pelagic bivalve species are relatively long ranging when compared with the aforementioned groups, they can also be useful as an independent tool in correlations.

3. Geological setting, lithostratigraphy, biostratigraphy and sampling

3.1. The Mayerling section

This section is located approximately 30 km southwest of Vienna, in the vicinity of the village of Weissenbach near Mayerling (Fig. 2a). We decided to use the name 'Mayerling' (rather than Weissenbach) because of its well-known historical background. From a paleogeographical point of view, this section belongs to the Northern Calcareous Alps (NCA), which are classically considered as a part of the southern Tethyan continental margin [13,14]. Paleomagnetic investigations in this area clearly suggest that a large-scale clockwise rotation occurred during the tectonic emplacement of the NCA with respect to Eurasia [15–18], the last phase of deformation probably being of late Eocene age [19,18].

The section comprises two sites sampled 750 m from each other. The first site is along a railway cutting (1, Fig. 2a) and the second is located on the two edges of an active quarry (2, Fig. 2a). In both sections the beds dip to the southeast, at approximately 55° in the railway cutting and at 40° in the quarry. The section consists of 25 m of whitish fine-grained pelagic limestones without major changes in the lithology. The biostrati-

graphic control is based on conodonts. Seventeen samples were taken throughout the section and indicated a lowermost Carnian age (Julian 1 zone) for the entire section. By comparison with the Bolücektasi Tepe section, where the lower Carnian is condensed (the Julian 1 zone is 3.5 m thick), the sedimentation rate here was about seven times higher.

We collected 67 minicores using a portable petrol-powered drill. the samples were oriented with a magnetic compass. Using the assumption of an equal duration of biochronological zones the duration of the sampled sequence is about 1.5 my, providing a mean time resolution of about 25 ky/sample. This resolution is twice that obtained from the BT section. The assumption for estimating the age of the Triassic stage boundaries has already been discussed in Gallet et al. [1].

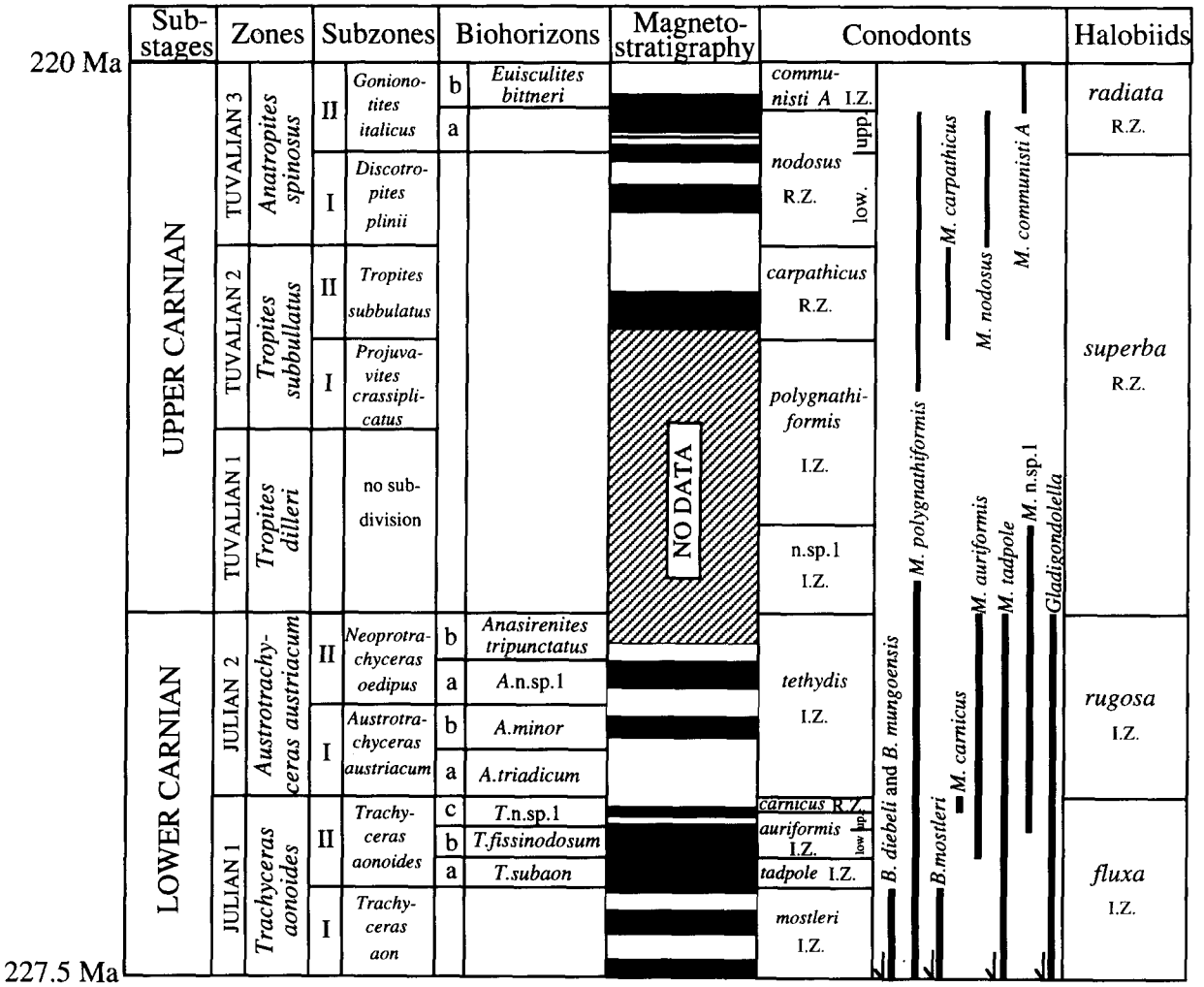


Fig. 1. Biochronology and magnetostratigraphy of the Carnian. The lower Carnian is defined by two ammonoid zones (Julian 1 and 2) and the upper Carnian by three (Tuvalian 1–3). This standard ammonoid zonation has been calibrated with the conodont and halobiid zones [Krystyn, in prep.]. Ten conodont zones are recognized throughout the Carnian, five in the lower Carnian and five in the upper Carnian. We have also indicated the composite magnetic polarity sequence determined from our results.

3.2. The Erenkolu Mezarlik section

The sampled section is located close to the village of Kemer in the western Taurides (southwestern Turkey, Fig. 2b) and about 3 km from the BT section. Both sections belong to the Barkirli Dag unit, which is part of the Antalya calcareous nappe system [21–24]. Previous results obtained from the BT section indicate that this unit was deposited on the northern margin of Gondwana, probably close to Arabia [1,25]. Analysis of declinations has provided two different directions, differing by about 30° on both sides of a major sedimentary gap during the middle Carnian. This difference is interpreted as a major episode of breakup of the northern part of Gondwana. The lithological succession at EM compares well with that at BT. This sequence of Hallstatt facies consists of about 5 m of pelagic limestones (about half the thickness of the Carnian BT sequence). The trends of the fold axes are close to those of the BT, and the succession is overturned, allowing a fold test that compares with the results from the BT section. The lithological descriptions are found in Marcoux [26] and Gallet et al. [1]. In this study, we have only sampled the Carnian sequence corresponding to the 'nodular' member (lower Carnian) and the 'bedded' member (upper Carnian). The biostratigraphy is based both on conodont and ammonoid zonation, and 25 samples have been collected for studying the biostratigraphy. The age of the sampled section ranges

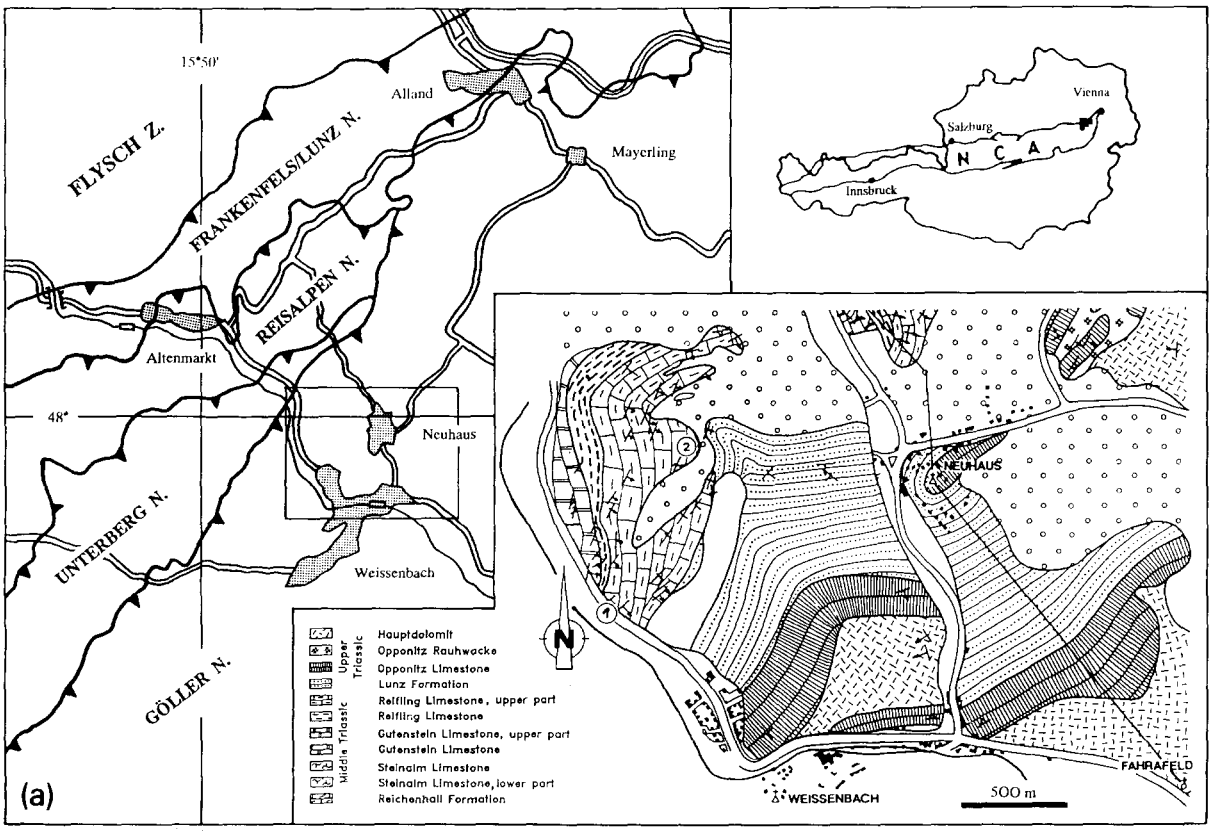
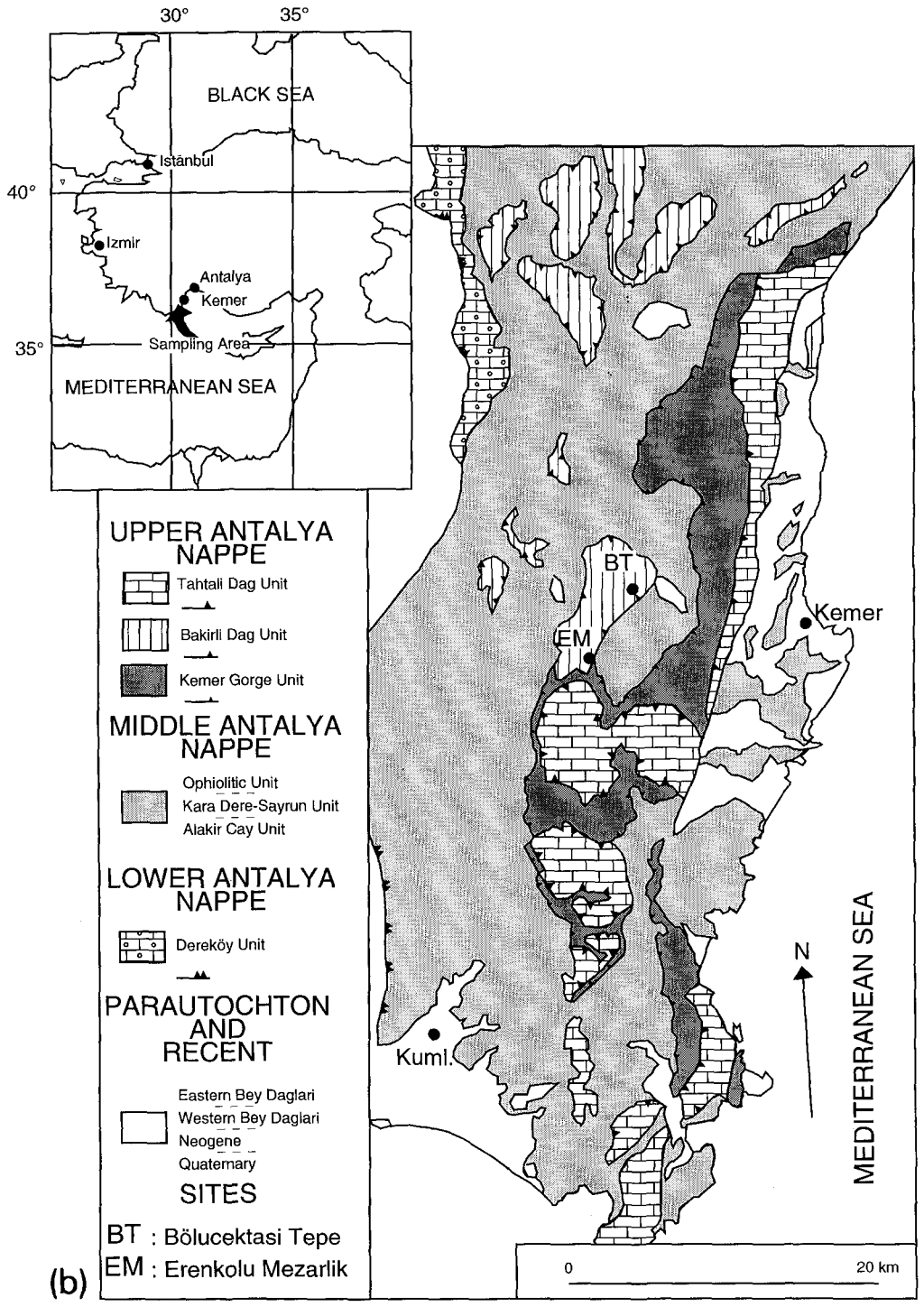


Fig. 2. Location map of the Mayerling (a) (after [20]) and Erenkolu Mezarlik (b) (after [24,25]) sections, and simplified geological and structural maps of the two sampling areas.


30°
35°
BLACK SEA
Istanbul
40°
Izmir
Antalya
Kemer
35°
Sampling Area
MEDITERRANEAN SEA
UPPER ANTALYA NAPPE
Tahtali Dag Unit
Bakirli Dag Unit
Kemer Gorge Unit
MIDDLE ANTALYA NAPPE
Ophiolitic Unit
Kara Dere-Sayrun Unit
Alakir Cay Unit
LOWER ANTALYA NAPPE
Dereköy Unit
PARAUTOCHTON AND RECENT
Eastern Bey Daglari
Western Bey Daglari
Neogene
Quaternary
SITES
BT : Bölucektasi Tepe
EM : Erenkolu Mezarlik
(b)
BT
EM
Kemer
Kuml.
N
MEDITERRANEAN SEA
0
20 km

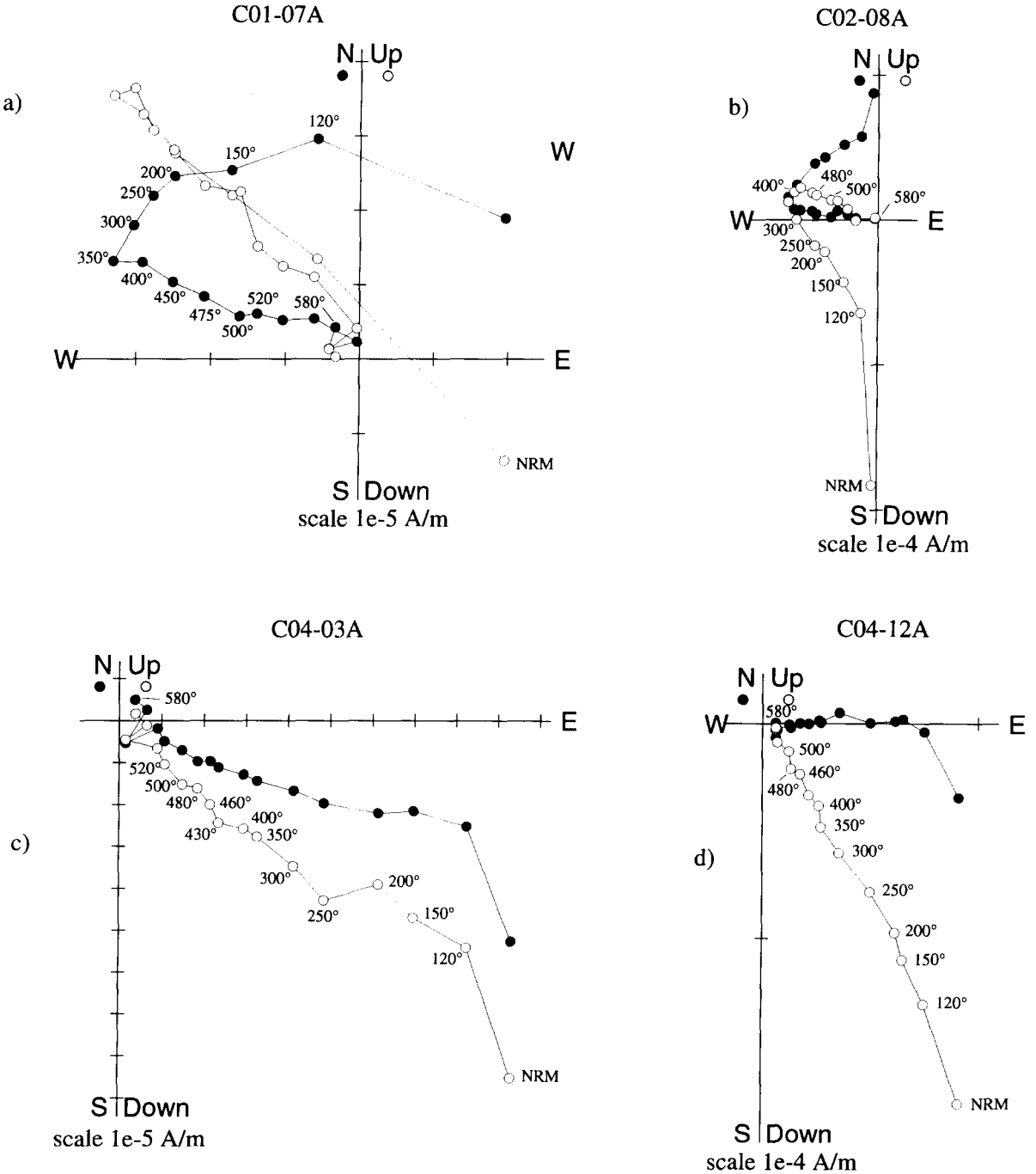


Fig. 3. Orthogonal vector diagrams of progressive thermal demagnetizations of samples from the Mayerling section. The closed/open symbols refer to the horizontal/vertical plane. The diagrams, which are in stratigraphic coordinates, clearly show high unblocking temperature components in normal and reversed polarity.

from the upper part of the Julian 1 zone (lower Carnian) to the Lacian 1 zone (lowermost Norian). Several sedimentary gaps are deduced from the biostratigraphic determinations. From base to top, a first gap is found at the Julian 1/2 boundary, a second at the Julian 2/Tuvalian 2 bound-

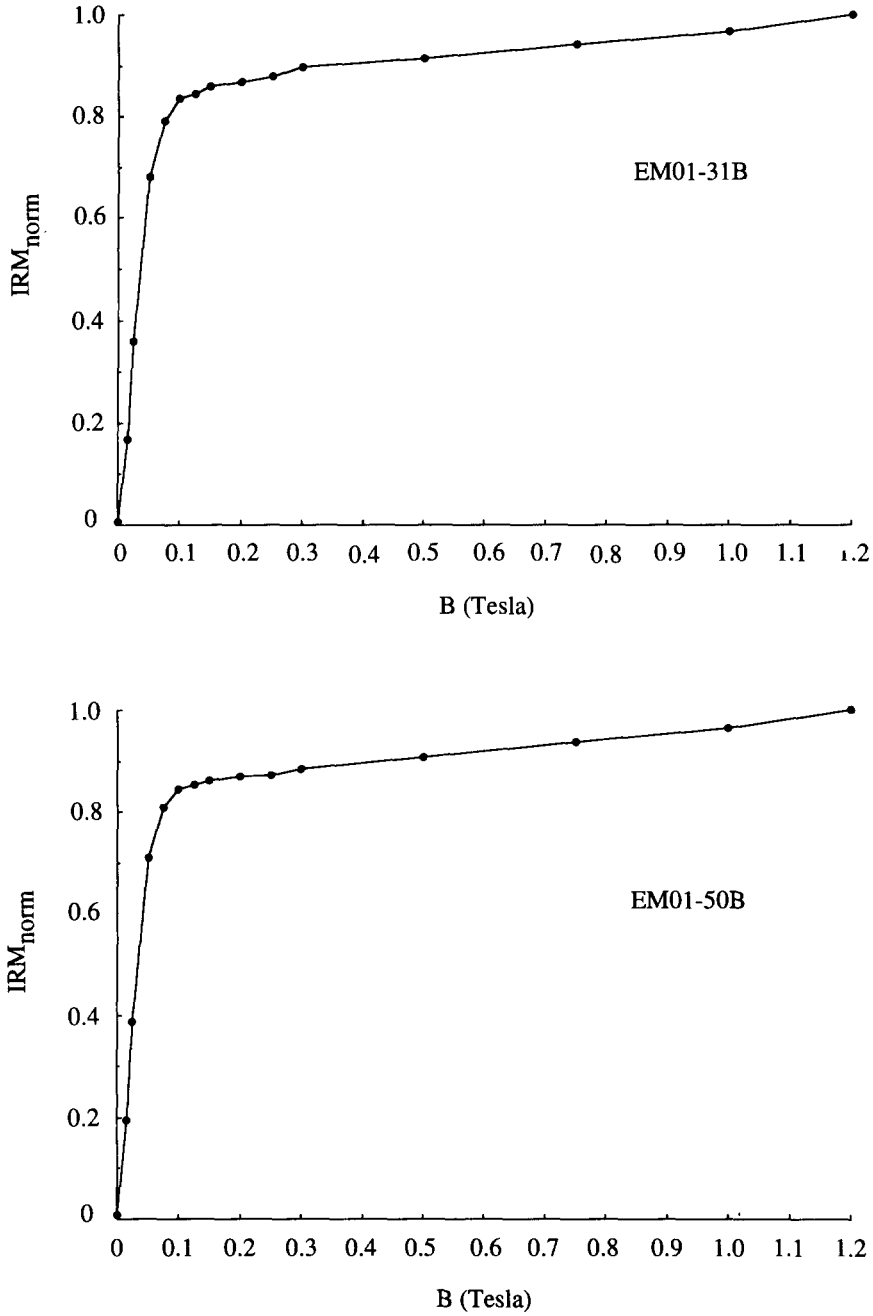


Fig. 4. IRM experiments up to 1.2 T conducted on two samples from the Mayerling section. A first mineral fraction is saturated in a 0.15–0.2 T field which may correspond to a mineral in the magnetite family. Moreover, a large fraction of a high-coercivity mineral is also observed.

ary (the Tuvalian 1 and the lower part of the Tuvalian 2 are missing) and the last at the Carnian (Tuvalian 3)/Norian (Lacian 1) boundary. As discussed in Théveniaut [25], the second gap would correspond to a major tectonic phase visible both in the EM and BT. However, the biostratigraphic data indicate that the sedimentary gap is shorter in the EM section, where the sedimentation resumed within the Tuvalian 2/II zone instead of the Tuvalian 3/II zone in the Bolücektasi Tepe section.

Fifty-three paleomagnetic samples have been drilled throughout the section, at a close spacing of about 10 cm. This sampling provides a mean time resolution of ca. 100 ky/sample. The resolution is thus half that reached in the Carnian BT sequence, and one fourth of that in Mayerling.

4. Paleomagnetic results and analyses

Magnetic measurements were carried out with a cryogenic CTF three-axis magnetometer in the magnetically shielded room at the *Institut de Physique du Globe* in Paris. The samples were thermally demagnetized in about 15–17 steps, using a laboratory-built furnace. Several samples were also demagnetized by alternating field (AF) with a GSD1 Schönstedt demagnetizer. The demagnetization data are plotted on orthogonal

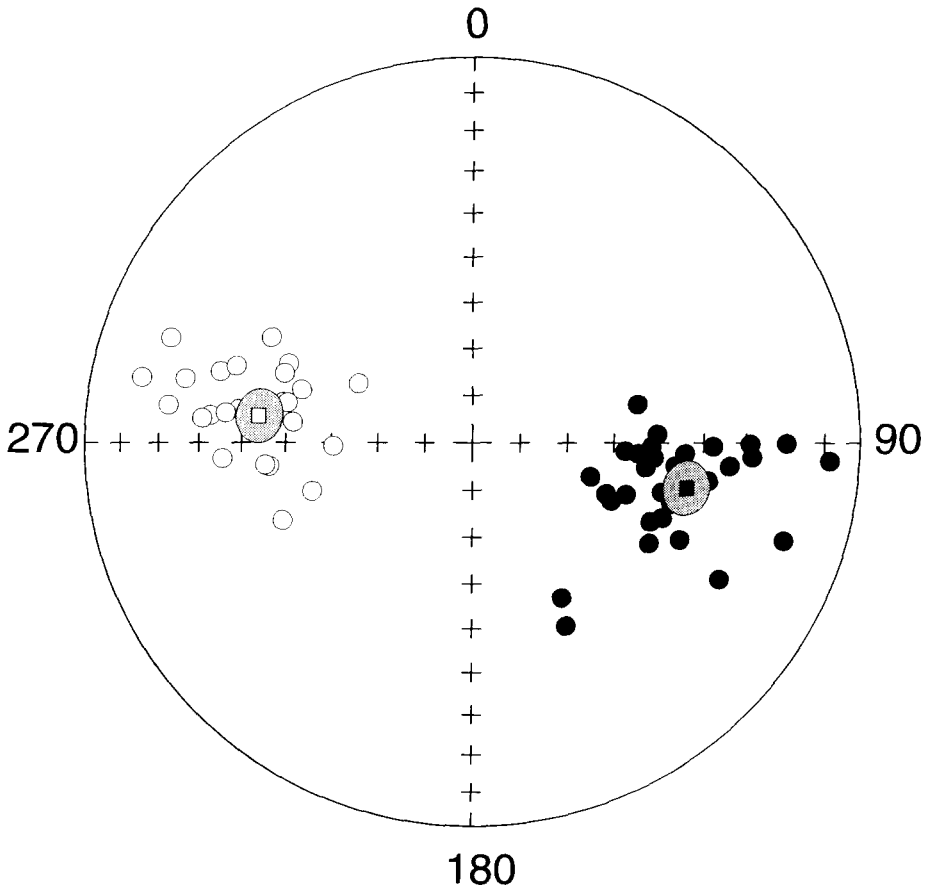


Fig. 5. Equal-area projection of the directions of the characteristic component obtained from the Mayerling section (after bedding correction). Two groups of antipodal directions are clearly observed. The closed/open square indicates the normal/reversed mean direction.

Table 1
Mean directions of the characteristic magnetic components obtained throughout the Mayerling section

	Component	N samp.	Decl. before tilt correction	Incl. before tilt correction	Decl. after tilt correction	Incl. after tilt correction	a_{95}	K
Mayerling (48.00°N,15.50°E)	Normal	37	30.1°	64.4°	102.0°	43.7°	5.3°	20.5
	Reversed	26	231.4°	-66.6°	277.2°	-43.5°	5.3°	29.3
	General	63	38.6°	65.7°	100.0°	43.6°	3.8°	23.5
	VGP	Lat.: 12.4°N Long.: 81.1°E dp: 3.0°, dm: 4.7° Paleolat.: 25.5°N						

vector diagrams [27] and the paleomagnetic directions were determined for each sample by principal component analysis [28], and by great circle analysis in some cases [29]. The mean directions were estimated using the statistical methods of Fisher [30] and McFadden and McElhinny [31].

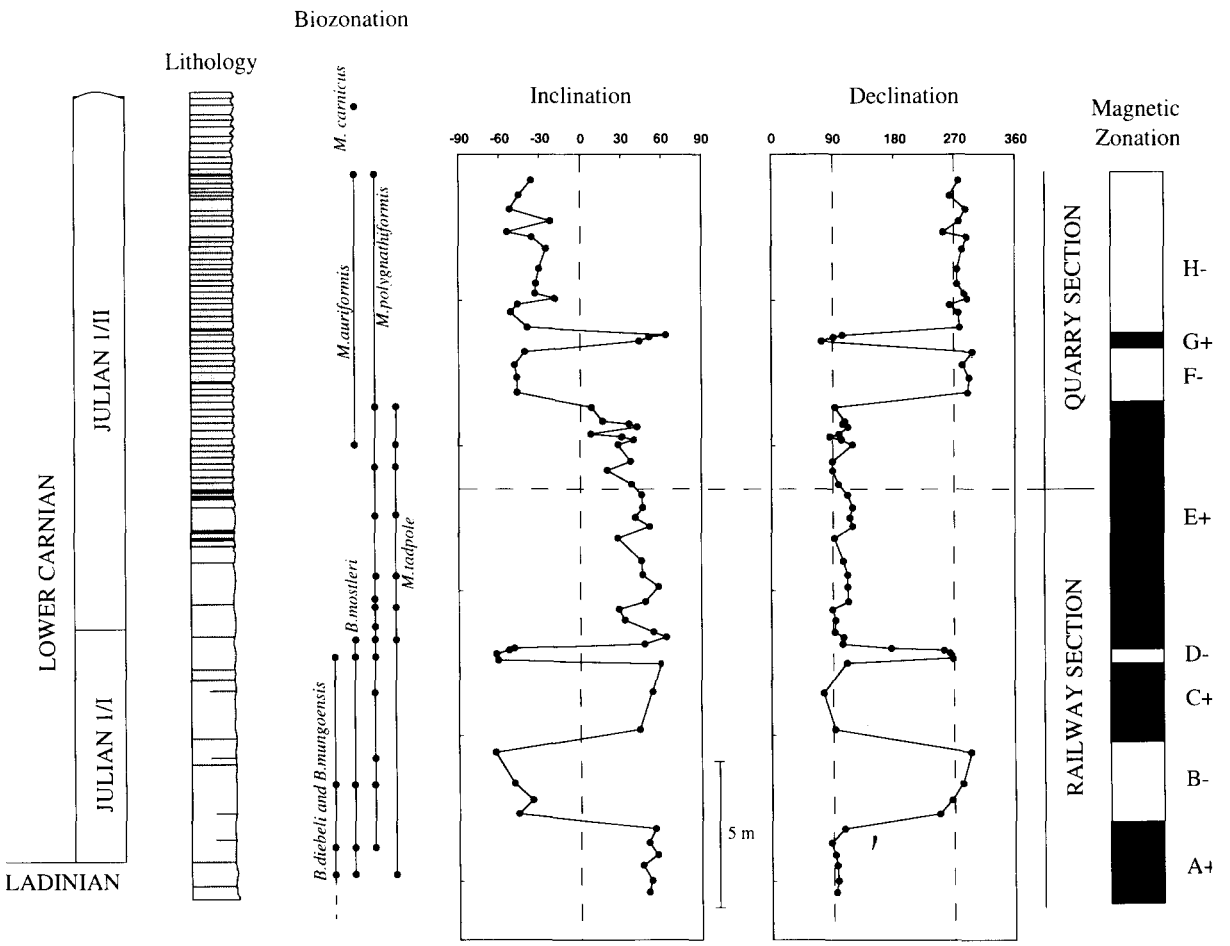
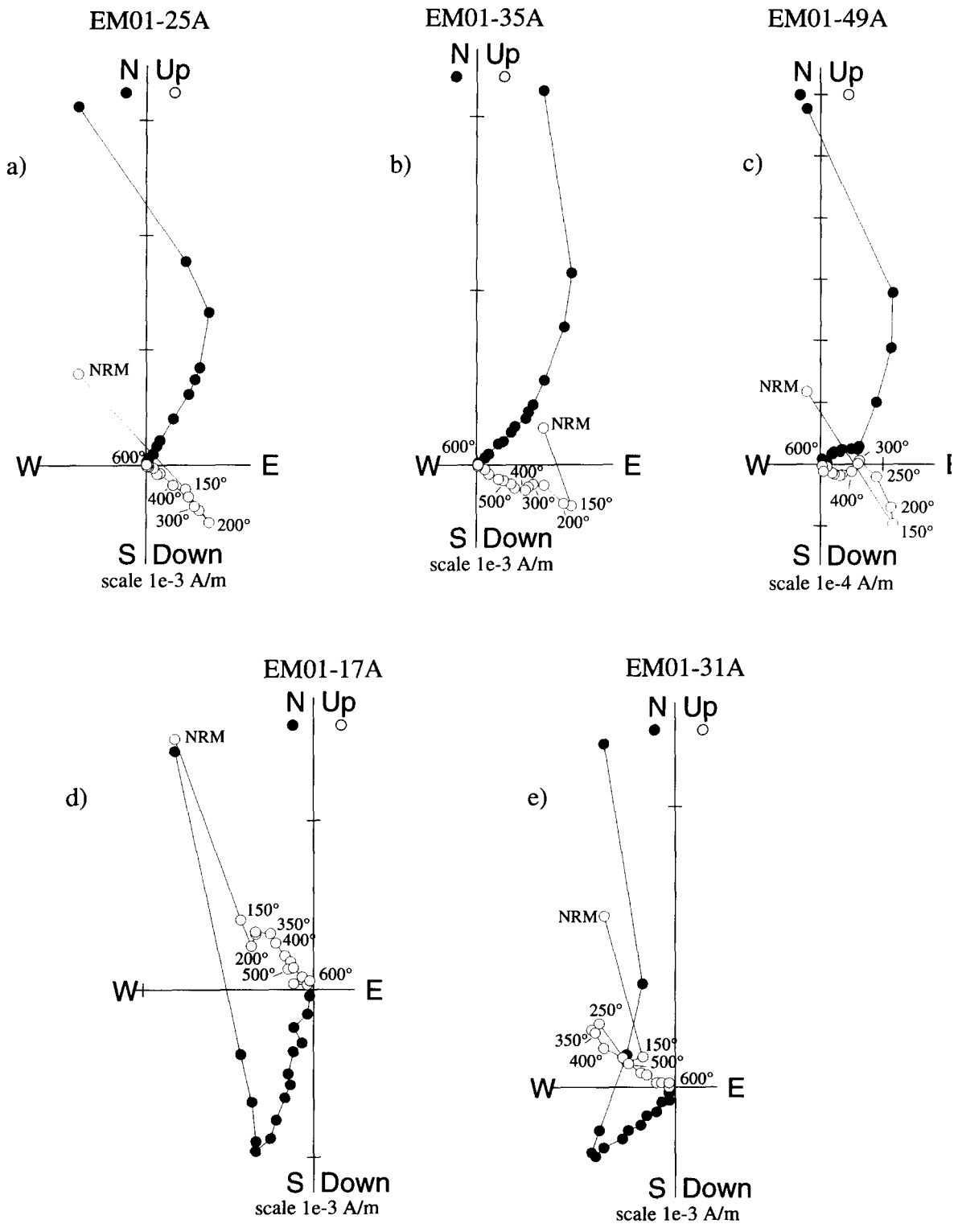


Fig. 6. Magnetostratigraphy obtained from the Mayerling section. A sequence of eight magnetic intervals correlated with the biostratigraphic zonation based on conodonts is observed. Each magnetic interval is defined by at least three samples. Following Alvarez et al. [32], the polarity intervals are designated by letters in ascending alphabetic order.


EM01-25A
a)
N Up
NRM
600°
W
E
150°
400°
300°
200°
S Down
scale 1e-3 A/m
EM01-35A
b)
N Up
NRM
600°
W
E
400°
500°
300°
150°
200°
S Down
scale 1e-3 A/m
EM01-49A
c)
N Up
NRM
600°
300°
W
250°
400°
200°
150°
S Down
scale 1e-4 A/m
EM01-17A
d)
N Up
NRM
150°
350°
400°
200°
500°
600°
W
E
S Down
scale 1e-3 A/m
EM01-31A
e)
N Up
NRM
250°
350°
150°
400°
500°
600°
W
E
S Down
scale 1e-3 A/m

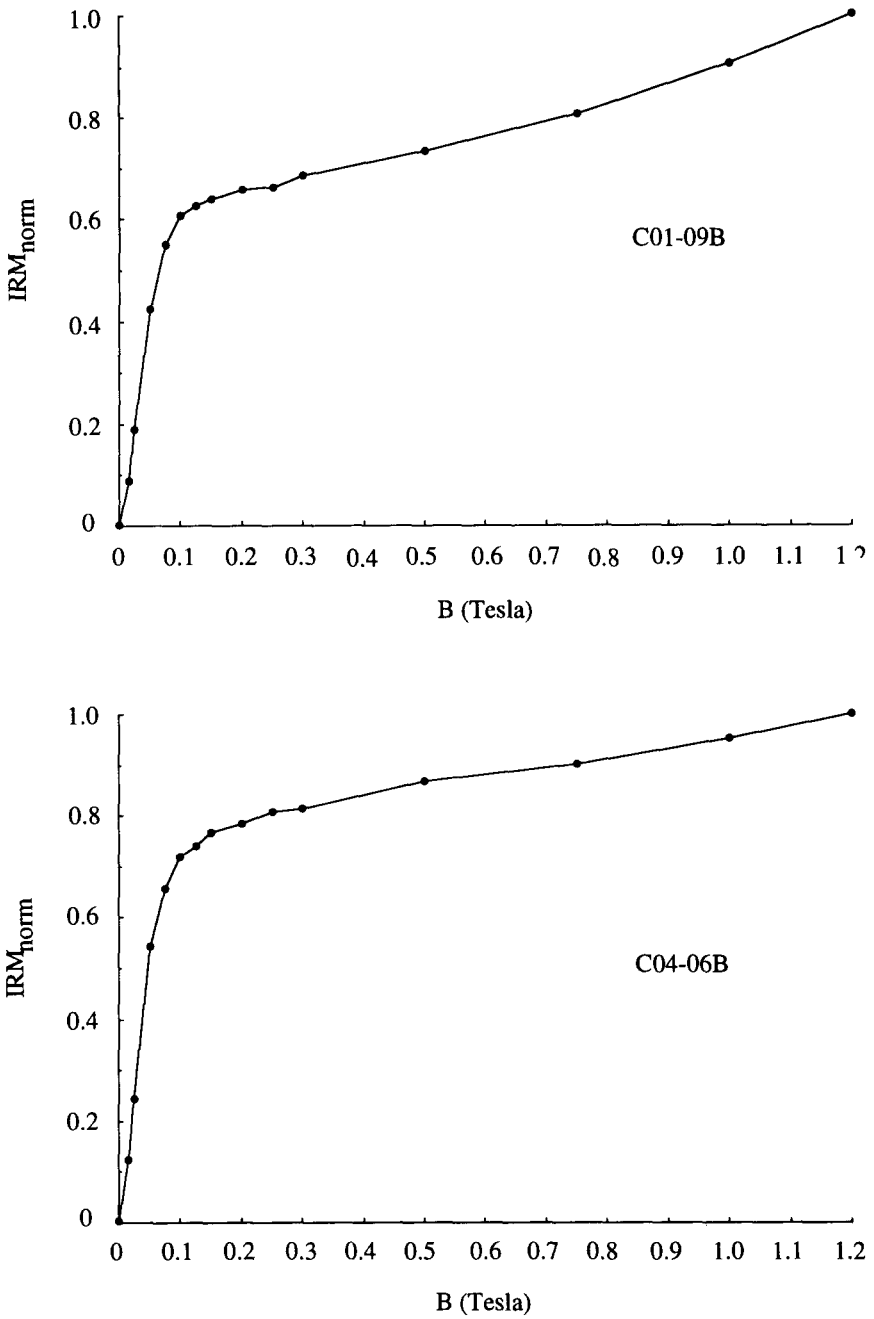


Fig. 8. IRM analyses on two samples from the Erenkolu Mezarlik section. The magnetization is mostly saturated in a 0.15 T field, which is likely to indicate a magnetization dominated by a mineral in the magnetite family. A small fraction of a high-coercivity mineral is also evident.

Fig. 7. Thermal demagnetization of samples from the Erenkolu Mezarlik section in stratigraphic coordinates (same convention as in Fig. 3). Three components are obtained. The first component is isolated in the low unblocking temperature range and has the present-day field direction. An intermediate component of reversed polarity is also isolated in the middle temperature range but its direction is never well isolated due to overlaps with the other components. Finally, a high unblocking temperature component is observed with the two polarity states (normal (a–c) and reversed (d and e)).

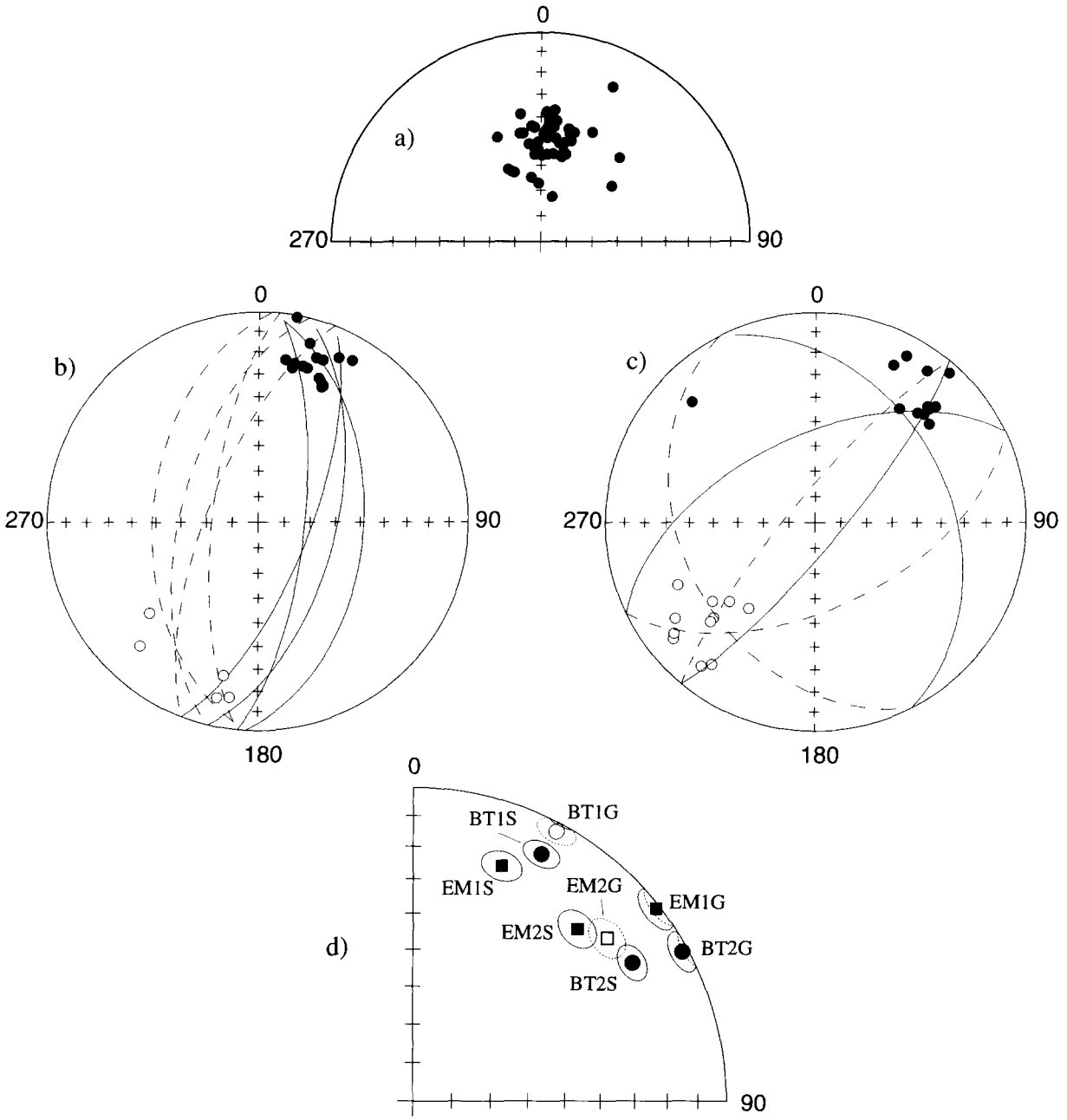


Fig. 9. Equal-area projection of directions of the low unblocking temperature component (a) (in geographic coordinates) and of the high unblocking temperature component (b and c) (in stratigraphic coordinates). The low unblocking temperature component clearly has the direction of the present-day field. Two different characteristic components are obtained (1) from the lower Carnian (b) and (2) from the upper Carnian (c). In particular, a noticeable difference is observed in declination (see also Table 2). The directions of Carnian age obtained from EM and BT are plotted together in (d). EM1G/S indicates the directions obtained from EM in geographic (G) and stratigraphic (S) coordinates for the lower Carnian; EM2G/S indicates the same for the upper Carnian. BT1G/S indicates the directions obtained from BT in geographic (G) and stratigraphic (S) coordinates for the lower Carnian; BT2G/S indicates the same for the upper Carnian.

4.1. The Mayerling section

The general thermal demagnetization characteristics show that an initial component is removed in the first demagnetization steps (i.e., < 150°C; Fig. 3). In the middle temperature range, up to approximately 400°C, the vector diagrams show large changes in direction. These changes indicate overlaps between components, one being a high unblocking temperature component which is clearly isolated between about 400°C and 600°C. This component may be either of reversed (Fig. 3a and b) or normal (Fig. 3c and d) polarity. In the highest temperature steps, between about 560°C and 600°C, the directions are more scattered, which may indicate that changes in the magnetic mineralogy occurred in the samples during the thermal treatment, although no noticeable change in low field susceptibility is observed.

IRM experiments conducted up to 1.2 T show that a first fraction of a magnetic mineral is saturated around 0.15–0.2 T and probably belongs to the magnetite family (Fig. 4). A fraction of a high-coercivity mineral is also clearly observed. The large drop in intensity at 120°C often observed after heating indicates that the high-coercivity mineral is probably goethite. Hematite, if present, cannot be characterized by unblocking temperatures greater than 600°C. Demagnetization in AF up to 80 mT does not provide further evidence of high-coercivity minerals due to the acquisition of spurious anhysteretic remanence (ARM) from about 40 mT onward.

We have plotted in Fig. 5 the tilt-corrected characteristic directions isolated in the samples from the Mayerling section. Two groups of directions are clearly observed and Fisherian means were computed both for the normal and the reversed components (Table 1). In stratigraphic coordinates, the two mean directions are statistically antipodal (open and close squares in Fig. 5) and the general mean direction, estimated from 63 samples, has an easterly declination (100.0°). This is in agreement with most of the Jurassic and Triassic directions already known from the Northern Calcareous Alps [e.g., 15–18]. The characteristic directions obtained from the quarry and from the railway sections define a magnetic zonation of eight intervals, each established by at least three samples (Fig. 6).

4.2. The Erenkolu Mezarlik section

Thermal demagnetization indicates three successive components (Fig. 7) identical to those observed in the nearby BT section. A first component of normal polarity is isolated in the low unblocking temperature range, between the NRM and about 200°C. Then, an intermediate component is obtained in the middle blocking temperature range, but its direction is never well isolated due to a significant overlap with the other components. This component, which has a reversed

Table 2
Mean directions of the characteristic components obtained throughout the Erenkolu Mezarlik section

Erlinkolu Mezarlik (36.65°N,30.55°E)	Component	N samp.	Decl. before tilt correction	Incl. before tilt correction	Decl. after tilt correction	Incl. after tilt correction	a_{95}	K
lower Carnian	Normal	15	52.1°	3.1°	19.0°	20.8°	4.7°	67.5
	Reversed	9	232.0°	2.7°	204.4°	-22.8°	10.8°	25.0
	General	24	51.7°	1.4°	20.7°	21.1°	4.5°	45.0
	VGP	Lat.: 58.2°N Long.: 169.3°E dp: 2.5°, dm: 4.7° Paleolat.: 10.9°N						
upper Carnian	Normal	11	47.0°	-16.1°	39.8°	21.5°	6.4°	51.5
	Reversed	13	232.7°	24.1°	227.5°	-28.3°	7.3°	33.5
	General	24	50.0°	-20.3°	43.7°	25.2°	4.8°	37.3
	VGP	Lat.: 44.5°N Long.: 139.9°E dp: 2.8°, dm: 5.2° Paleolat.: 13.2°N						

polarity in stratigraphic coordinates, affects the demagnetization path to different degrees (compare samples EM01-25A (Fig. 7a), EM01-35A (Fig. 7b) and EM01-49A (Fig. 7c)). Finally, in most samples a high unblocking temperature component of normal (Fig. 7a–c) and reversed (Fig. 7d and e) polarity is observed between about 475°C and 580–600°C. In some cases, the high-temperature component could not be successfully isolated by principal component analysis and only great circles containing this component could be determined [29].

IRM experiments up to 1.2 T indicate that the magnetization is dominated by grains saturated in a 0.15 T field, although a high-coercivity mineral is also observed (Fig. 8). Again, this behaviour is similar to the results obtained from the BT section [1]. Therefore, these data suggest that the magnetization is essentially carried by a mineral of the magnetite family and that a small fraction of goethite and hematite, with a low unblocking temperature, exists in the samples from EM.

The directions obtained for the low unblocking temperature (i.e., < 200°) component are plotted in Fig. 9a in geographic coordinates. This component clearly has the present-day field direction. As in BT, two different high unblocking temperature components are observed in EM. For this reason, we divided the EM section into two subsections on either side of the sedimentary gap

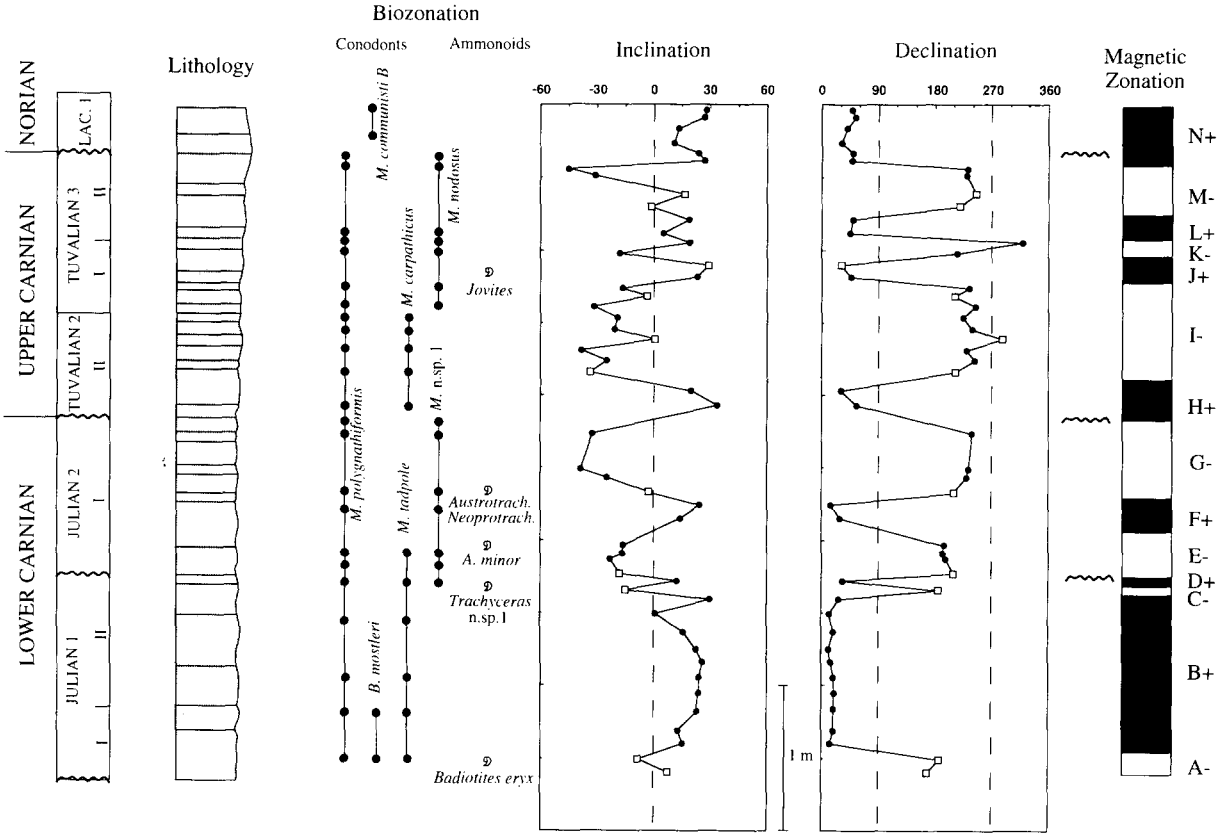


Fig. 10. Magnetostratigraphy and biostratigraphy obtained from the Erenkolu Mezarlik section. The biostratigraphy is based both on ammonoids and conodonts. Wavy lines indicate sedimentary gaps. Fourteen magnetic intervals are observed. Three of these are defined by one sample and the reversed polarity interval at the base of the section (Julian 1/I zone) is only defined by two great circles containing the characteristic component. The open squares indicate biased directions obtained from the last demagnetization data in the samples for which only great circles could be determined. The magnetic intervals are designated by letters [32].

including the lower part of the late Carnian. Therefore, the first subsection contains the lower Carnian (Julian 1 and 2 zones, Fig. 9b) and the second contains the upper Carnian (Tuvalian 2 and 3 zones, Fig. 9c). The mean directions were computed using the McFadden and McElhinny

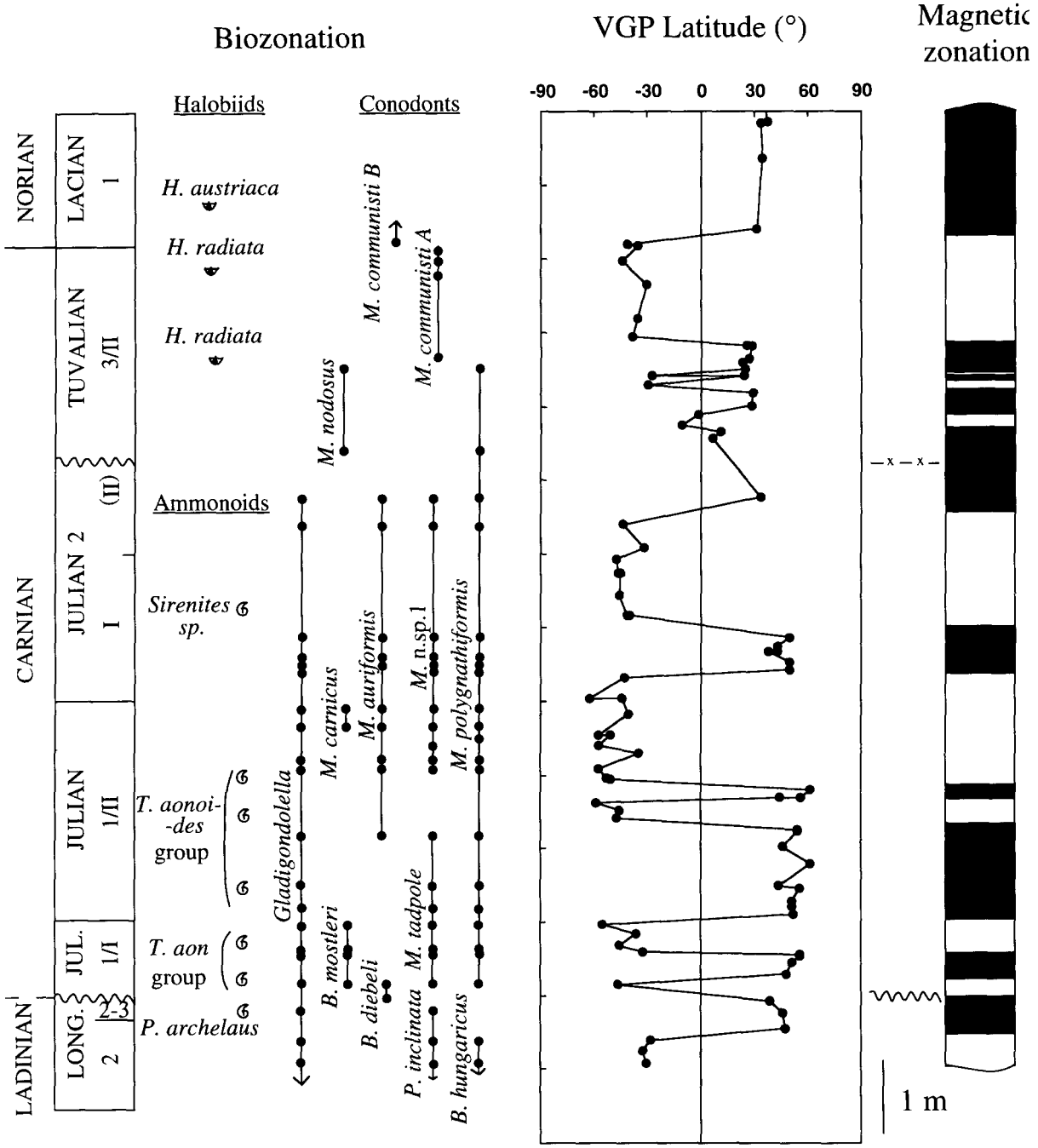


Fig. 11. Magnetostratigraphy and biostratigraphy obtained from the Carnian part of the Bolücektasi Tepe section [1]. Additional biostratigraphic data based on conodonts have been considered.

[31] method (Table 2). In both subsections these estimates show that the directions of normal and reversed polarity are antipodal and two 'general' mean directions were confidently computed for the lower Carnian and the upper Carnian. A noticeable difference in declination is observed between the two parts of the EM section ($dD = 23.0 \pm 6.7°$). This difference is statistically similar to that observed in the BT section ($dD = 28.3 \pm 4.0°$, [1]) and suggests a significant rotation at the 95% level of the Bakirli Dag unit during the Carnian [25]. The directions obtained from EM and BT can be used to perform a fold test (Fig. 9d). Because only two sites are available, a statistical fold test is not significant at the site level. However, for both periods, a positive fold test is obtained at the sample level when only the inclinations are considered (lower Carnian, $K_s/K_g = 1.23$, $N = 40$; upper Carnian, $K_s/K_g = 1.24$, $N = 75$) [33]. Together with the presence of normal and reversed polarity antipodal directions in the EM section, the isolated characteristic magnetization can be confidently considered as primary.

The characteristic directions obtained within Erenkolu Mezarlik define a magnetic zonation containing fourteen intervals (Fig. 10). Due to the low sedimentation rate, three of these intervals are defined by only one sample. Moreover, in

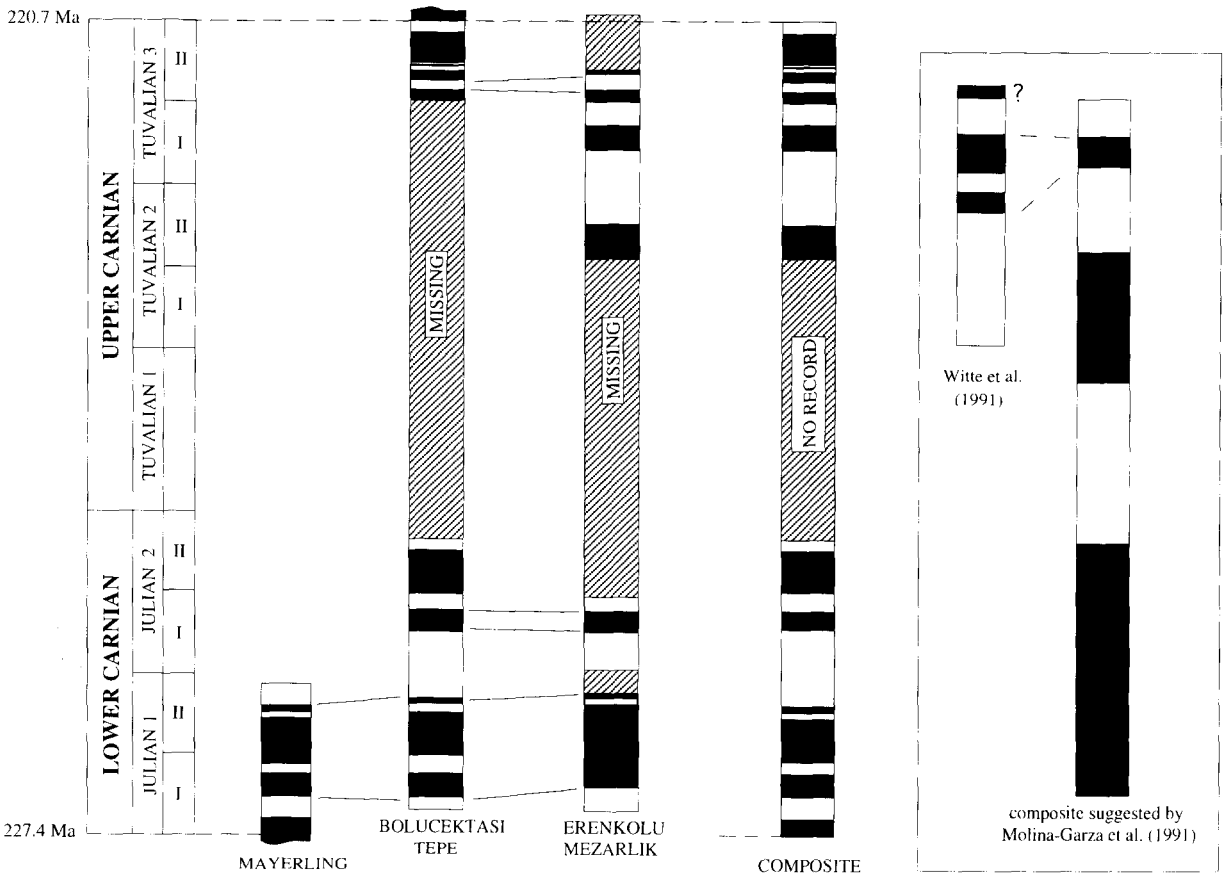


Fig. 12. Synthesis of the magnetostratigraphic results for the Carnian obtained from the Mayerling, the Erenkolu and the Bolücektasi Tepe sections. The numerical time frame has been estimated using the alternative concept of equal duration for the ammonoid zones [1]. A correlation is proposed between these three sections (continuous lines) and a composite Carnian magnetic polarity sequence is proposed although the lower part of the upper Carnian remains undocumented. Our composite sequence is compared to the available Carnian magnetostratigraphic results obtained from North America summarized by Witte et al. [7] and Molina-Garza et al. [5].

some samples, the characteristic component cannot be properly isolated because the directions lie along great circles. In these cases, only biased estimates of the 'true' directions, indicated by the open squares in Fig. 10, can be obtained from the last demagnetization data points. Of course, these rough estimates, which are particularly biased in inclination, were only considered to draw Fig. 10, and were not used to compute the mean directions.

5. Discussion

The Mayerling and the EM sections provide parts of the Carnian magnetic polarity time scale that we can compare with the Carnian magnetic sequence previously obtained in the BT section. In Fig. 11, we recall this latter magnetic sequence considering the previous and new biostratigraphic (conodont) data. Seventeen magnetic intervals are observed in this part of the section [1]. Furthermore, the magnetostratigraphic study of a nearby small section (about 50 m from the main BT section) showed the presence of an additional reversed interval in the upper part of the Julian 2/II zone (lower Carnian) that is missing in the main BT section due to faulting [25] (see Fig. 12). The comparison between the different sequences is shown in Fig. 12 using a linear time scale derived from the approximation of an equal duration for the Triassic biochronological zones [1]. Such an assumption is intended to palliate the poor reproducibility of the radiometric ages of the Triassic stage limits [34–36]. Based on biostratigraphic control, comparison between Mayerling and BT shows a satisfactory polarity correlation (continuous lines in Fig. 12). The magnetostratigraphy from the Mayerling section extends the previous Carnian magnetic sequence to the Ladinian/Carnian boundary, which was formed during a normal magnetic interval. This result is in agreement with the magnetostratigraphic sequence obtained from the Gamsstein section [3]. It is also worth noting that no additional short magnetic interval with respect to BT is observed within the Mayerling section, although a large difference in sedimentation rates exists between these two sections. A satisfactory correlation is also obtained between the BT and EM sections (Fig. 12). Due to a lower sedimentation rate, at least one short polarity interval is missing in EM, in the Julian 1/I zone. However, the EM section provides additional magnetic intervals in the Tuvalian 2 and 3/I zones, rocks of that age being absent in the BT section due to a wider sedimentary gap. By combining the three sections we propose a composite Carnian polarity sequence that contains 24 intervals (Fig. 12). This sequence is not complete because no well-dated magnetostratigraphic datum is yet available for the early part of the upper Carnian (Tuvalian 1 zone and the lower Tuvalian 2 zone). When we compare our sequence to the data previously obtained from the Newark basin [7] and to the composite sequence suggested by Molina-Garza et al. [5] no clear correlation is obtained (Fig. 12). In particular, a large discrepancy is observed in the lower Carnian where one normal interval, if we follow Molina-Garza et al. [5], would correspond to eleven intervals from the new sequence. This discrepancy probably results from different temporal constraints (linked to different sedimentation environments) and from different stratigraphic completeness and sampling densities between the data used for the two compilations. Finally, when the sequence that we propose in this study is interpreted in terms of magnetic reversal frequency, a value of about 4.5 reversals/my is estimated. This frequency estimate slightly decreases if the Harland et al. [34] and the Odin and Odin [36] time scales are considered (respectively about 3 and 3.5 reversals/my).

The correlation observed between the Mayerling and Bolücektasi Tepe magnetostratigraphic sequences has important implications for the origin of the Antalya nappes. Indeed, the inclinations determined in the BT and KT sections have, surprisingly, been found to be the opposite of each other [2]. A positive inclination may equally result from magnetization in the northern hemisphere during a normal polarity interval or from magnetization in the southern hemisphere during a reverse polarity interval. The situation is even more complicated in our case because both units are located in the Antalya nappes, so that large

rotations linked to the complex tectonic history of this area can also be expected. Such large rotations may lead to an additional hemisphere ambiguity. The polarity correlation between the overlapping part of the BT and KT sections during the Norian led us to propose that the two sections, which are now located about 150 km from each other, were deposited approximately 2500–3000 km apart in two different hemispheres [2]. Of course, the determination of the deposition hemisphere for one of the sections at least is required in our case in order to assess the correct magnetic polarity. Paleogeographic evidence [24] indicates that the BT section was deposited close to the northern tip of the Arabian promontory. Moreover, the clear northward shift in latitude between the Carnian and the Norian in the BT section is coeval with the general motion of Gondwana during the same time and supports a northern origin for the BT. We thus proposed in our earlier work that the BT sediments were deposited in the northern hemisphere and consequently those of the KT were deposited in the southern hemisphere. Additional verification is provided by the comparison between the Mayerling and BT sections. Indeed the Mayerling section is part of the Northern Calcareous Alps. This region has been studied extensively both by geologists and by paleomagnetists [15–18], and its northern origin appears to be beyond any doubt. The satisfactory correlation which is observed between the lower Carnian Mayerling and BT sections clearly identifies a northern hemisphere origin of the BT. Further magnetostratigraphic and paleomagnetic studies of Turkish sections are presently in progress, in order to decipher the complex origin of the Antalya nappes.

Acknowledgements

This research was financed by *DBT Instabilités* (contribution 697). The fieldwork by L.K. in Turkey in 1993 was funded by the Austrian National Commitee for the IGCP as part of IGCP Project 395. We thank C¸ Sengör for helpful support in the logistic organization of the fieldwork. We also thank H. Bucher and V. Courtillot for their critical reading of the manuscript. This is IPGP contribution 1320.

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